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Cybersecurity
PhD in Cybersecurity

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Chapter 1

Internet

1.0.1 What is internet

Internet is a network of networks. There are hierarcical networks such as:

- Internet backbone: connecting the ISPs' backbones
- ISP backbone: connecting organizations' backbones
- Organization backbone connects local area networks (LANs)
- LAN connects end systems

Internet is diveded in:

- Access network: connects end systems to the first router (e.g., DSL, cable, fiber, wireless)
- Physical media: carries signals between devices (e.g., twisted pair, coaxial cable, fiber optic, radio waves)
- Network core: connects the access networks (e.g., routers and links). Routers work togheter to figure a path for each packet from source to destination. The routing table are mantained in real time and a routing algorithm can adapt to changing internet conditons.

1.0.2 protocols

Protocols specify rules about the service, such as:

- procedure rules: types and sequence of messages exchanged, syntax and semantics, action to take in respect of messages and events
- message format size and coding
- timing to wait between events, timeouts, flow control

1.0.3 Access layer

Its constituted by network with end points of the same local management, provides connectivity among station on the same network and they can comunicate directly among them.

Ethernet(IEEE 802.3) is the most common access control protocol. Each host has a NIC (Network Interface Card) with a fixxed address. MAC addresses are 48bits long and uniquely identify the host in the network. Each ethernet frame has a fixed format.

Figure 1.1: Ethernet Frame Format

All the host connected to the same transmission system are a nework. An ethernet network costitutes a brodcast domani, where, ideally, frames sent in that domain are recived by all the host. Switches limit the explosion of the packets. Only the brodcast messages are replicated. Switches remembers the sourca MAC address on different ports. They only replicate the frame on the segment where the MAC address replies. Ethernet is inefficent in large network, as it uses brodcasst packets.

1.1 IP

1.1.1 IP Addresses

There are two version of IP: IPv4(32bit) and IPv6(128bit).

There are three types of IP addresses:

- Unicast: identifies a single host on the network
- Multicast: identifies a group of hosts on the network
- Broadcast: identifies all hosts on the network

there are classes of IP addresses:

- Class A: 24 bit for host.
- Class B: 16 bit for host.
- Class C: 8 bit for host.
- Class D: Used for multicast addressing.
- Class E: Reserved for future use.

There are routable and non routable IP addresses. The routable have to be unique on the internet, whike non routable can exist in different network but can not be used on internet.

1.2 Layering

the comunication between hosts in the network is organized in task assigned to a layer. Each layer offers a service to the layer above and uses the services of the layer below. the data is trasfered from the application level to application level over a network. Each layer adds protocol information and data to the layer below. The physical layer send the data physically to the destination. In the destination, the phisical layer sends the data up the stack. Each protocol reads the appropriate protocol information ad forward the data to the layer above.

There are 2 main layering models:

- ISO/OSI model: 7 layers (Physical, Data Link, Network, Transport, Session, Presentation, Application)
- TCP/IP model: 4 layers (Link, Internet, Transport, Application)

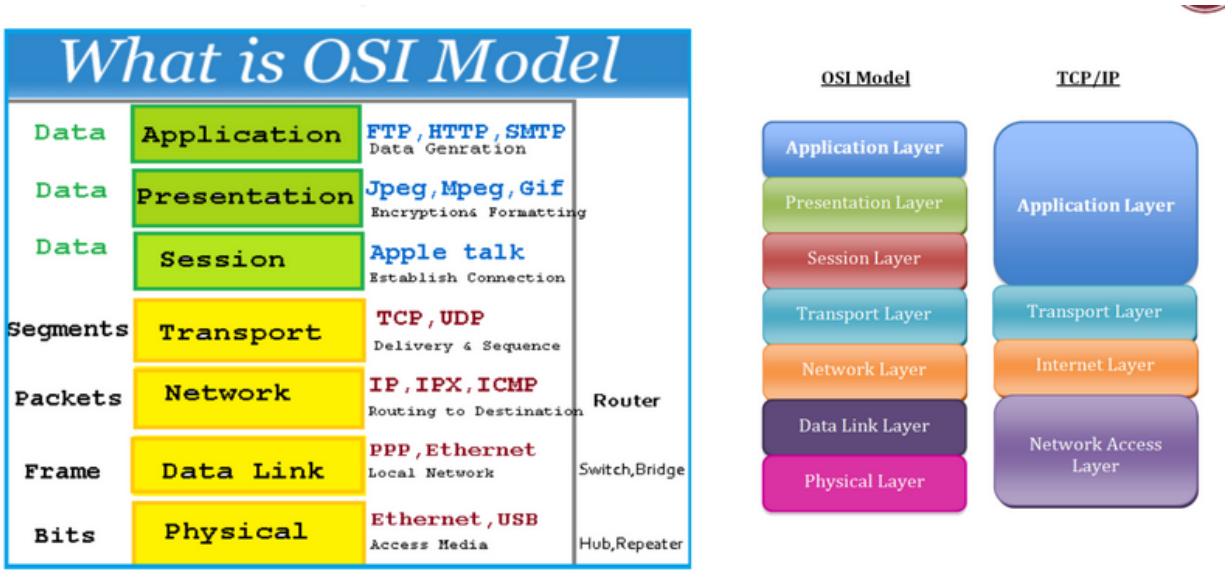

Figure 1.2: Models

1.2.1 TCP/IP model

- application layer: Corresponds to the top three layers of the OSI model.(Protocols: SMTP (sending e-mail), HTTP (web), FTP (file transfer), and others). Address: internet name (e.g., www.google.com)
- Transport layer: Equivalent to Layer 4 (Transport) of the OSI model.(Protocols: TCP, UDP). Address: port number (e.g., 80 for HTTP)
- Internet: Equivalent to layer 3 (network) of the OSI model.(Protocols: IP, ICMP, IPSec). Address: IP address (e.g., 192.168.1.1)
- Datalink: Equivalent to layer 2 (data link) of the OSI model.(Protocols: Ethernet, WiFi, ARP, etc.). Address: MAC address (e.g., 00:0a:95:9d:68:16)
- Physical layer: Equivalent to Layer 1 (Physical) of the OSI model.

Chapter 2

iPv6 addressing

2.1 IPv6 introduction

Developed in the 90s, written in exadecimal, has 128-bit addresses. It has stateless autoconfiguration, end-to-end reachability eithout using NAT, support for mobility, easy peer to peer networking.

iPv6 does not use NAT. NAT has been used to hide customers, but it creates some issues, like peer to peer networking.

iPv6 addresses are 128 bit repreented in hexadecimal and has the following format:

- 1 hex digit = 4 bit
- 8 groups of 4 hex digits between 0000 and FFFF
- groups are separated by colons
- leading zeros in a group can be ommited
- one or more consecutive groups of zeros can be replaced with a double colon (::)

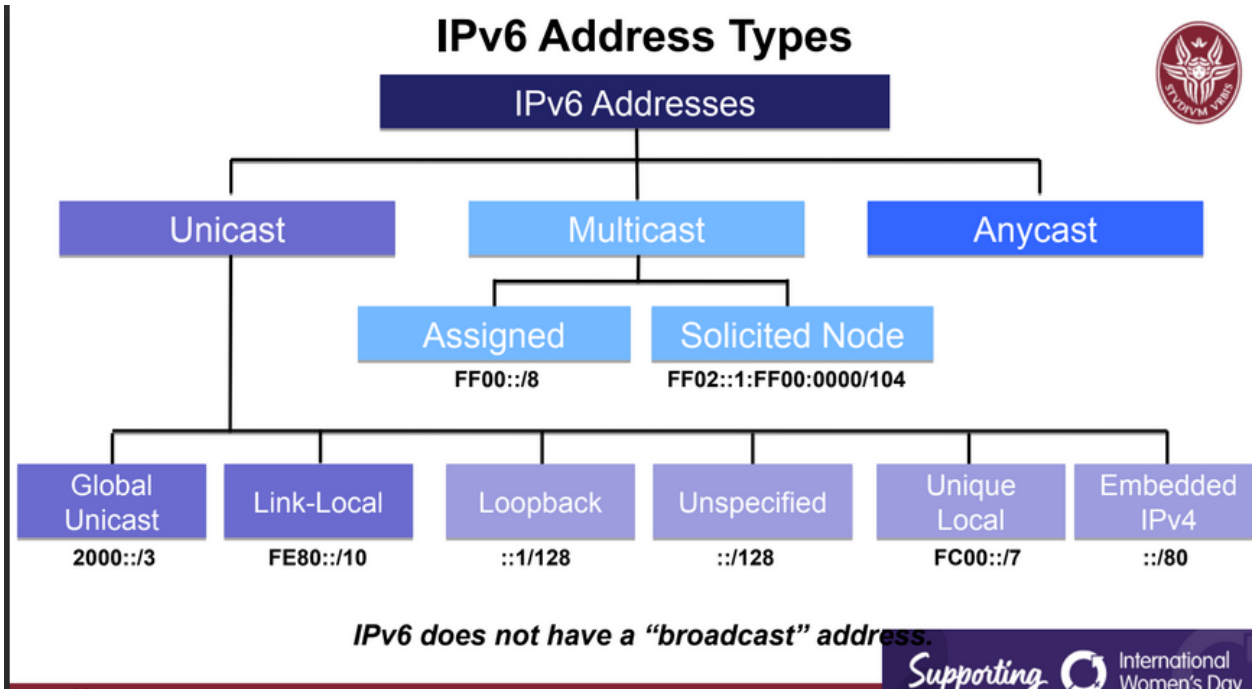

Figure 2.1: IPv6 address format

- IPv6 source: always a unicast (link local or GUA)
- IPv6 destination: unicast, multicast, or anycast
- Global Unicast Address (GUA): globally routable and reachable on the IPv6 internet. It has the following format: 2000::/3 (first 3 bits are 001). The next 45 bits are the global routing prefix, assigned by IANA to RIRs. The next 16 bits are the subnet ID, used to identify subnets within an organization. The last 64 bits are the interface ID, used to identify an interface on a subnet.
- prefix: equivalent to the network address of an IPv4 address
- prefix length: equivalent to the subnet mask of an IPv4 address
- interface ID: equivalent to the host address of an IPv4 address

2.1.1 Link-local addresses

Used to comunicate with other devices on the link. Not routable off the link, only have to be unique on the link. Not included in the IPv6 routing table. An ipv6 devise has at least one local link address.

The router link local is used as the default gateway. Routers exchange routing messages and use the link local address as next hop address in the routing

2.1.2 SLAAC

Stateless Address Autoconfiguration (SLAAC) allows a host to automatically configure its own IPv6 address without the need for a DHCP server. The host generates its own interface ID and

combines it with the network prefix obtained from the router advertisement messages to create a unique IPv6 address.

The idea is to rovide additional addresses with short lifetime and using themm as source address when originating connections. Same prefix as a ublic address, randomized value for interface ID. It is common to have multiple temporary addresses to make sure existing connections continue while a new temp address is created for a new connection.

2.1.3 ICMPv6 neighbor discover protocol

ICMPv6 Neighbor Discovery defines 5 different packet types:

- Router Solicitation Message
- Router Advertisement Message
- Neighbor Solicitation Message
- Neighbor Advertisement Message
- Redirect Message

When a host want an IPv6 address, it sends a Router Solicitation message to the all-routers multicast address. The router responds with a Router Advertisement message- There are 3 types of Router Advertisement messages:

- Option 1: SLAAC, No DHCPv6 "I'm everything you need (Prefix, Prefix-length, Default Gateway)"
- Option 2: SLAAC + Stateless DHCPv6 for DNS address "Here is my information but you need to get other information such as DNS addresses from a DHCPv6 server." (DNS can be in RA)
- Option 3: All addressing except default gateway use DHCPv6 "I can't help you. Ask a DHCPv6 server for all your information."

2.1.4 multicast addresses

Multicast addresses are used to send a packet to multiple destination simultaneously. There are two types of multicast addresses: Assigned and Solicited-node.

Multicast address have the prefix FF00::/8. they use a 4 bits field to define the range of the multicast packet, called scope:

- 0: reserved
- 1: interface-local scope (packets are only valid on the same interface)
- 2: link-local scope (packets are only valid on the same link
- 5: site-local scope (packets are only valid within the same site)
- 8: organization-local scope (packets are only valid within the same organization)

- E: global scope (packets are valid on the entire internet)

4 bits are used to define the flags:

- 0: permnent, well known multicast address
- 1: transient, dynamically assigned multicast address

Chapter 3

IPv4 vs IPv6

- version: Ipv4 contains 4, IPv6 contains 6
- internet header lenght(IHL): ipv4 has a 32 bit field including any options or padding, ipv6 has no IHL, his header is fixed at 40bytes
- ipv4 has "type of service" field, ipv6 has "traffic class" field
- ipv6 has "flow label" field to identify packets in common stream or flow, ipv4 has no equivalent
- ipv4 has 16 bit "total length" field, ipv6 has 32 bit "payload length" field
- ipv4 uses fragmentation, ipv6 does not use fragmentation, the source host is responsible for fragmentation
- ipv4 protocol, ipv6 next header
- ipv4 TTL, ipv6 hop limit
- ipv4 header checksum, ipv6 has no header checksum
- ipv4 has options and padding, ipv6 has extension headers. Next headers identify the protoco carried in the data portion of the packet

3.1 IPv6 security

IPv6 is not more secure than IPv4 and its security is built in. Even if there is no NAT in IPv6, it is possible to use firewalls to filter traffic and you are not exposed to attacks from the internet. IPv6 is resilient aganinst brute force scanning, but there are other scanning techniques.

There are lots of threats to IPv6, such as:

- IP spoofing: using a fake ip source address. Solution: ingress filtering, egress filtering, unicast reverse path forwarding (uRPF)
- using traffic clas or flow label to bypass security devices. Solution: firewall should inspect traffic class and flow label fields

- routhing header(Includes one or more IPs that should be “visited” in the path) attacks: using extension headers to bypass security devices. Solution: firewall should inspect extension headers
- fragmentation attacks: using fragmentation to bypass security devices. Solution: firewall should inspect fragments and reassemble them before inspection

3.2 associated protocols

3.2.1 ICMPv6

Provides error messages and informational messages. It is used for error reporting, diagnostics, and neighbor discovery. It is an integral part of the IPv6 protocol suite and is used for various functions such as path MTU discovery, router advertisement, and neighbor solicitation.

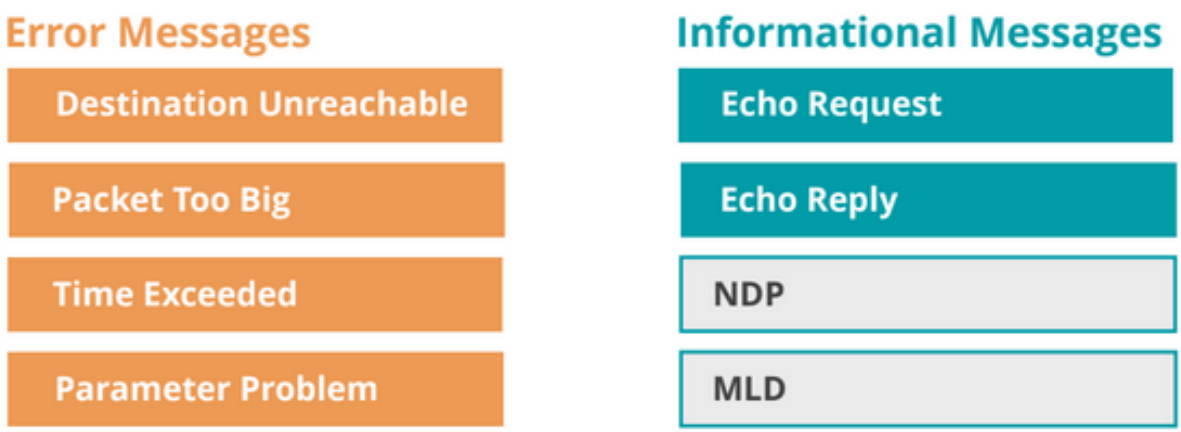

Figure 3.1: ICMPv6 message types

3.2.2 NDP

Used on a link to discover other devices, their link-layer addresses, and to maintain reachability information about the paths to active neighbors. It is used for address autoconfiguration, address resolution, neighbor unreachability detection, and router discovery.

NDP threats include:

- Neighbor solicitation/advertisement spoofing: an attacker can send fake neighbor solicitation or advertisement messages to redirect traffic or cause a denial of service. Solution: use secure neighbor discovery (SEND) which uses cryptographic methods to authenticate messages.
- Redirection/DOS attacks: an attacker can send fake redirection messages to redirect traffic or cause a denial of service. Solution: use SEND to authenticate messages.
- Man-in-the-middle attacks: an attacker can intercept and modify neighbor discovery messages to redirect traffic or cause a denial of service. Solution: use SEND to authenticate messages.

3.2.3 MLD

MLD (multicast listener discovery) is used on local link, uses ICMP, required by NDP. IPv6 nodes use it when joining a multicast group

Chapter 4

Traffic regulations

4.1 Firewalls

A firewall is a device that acts like a router and implements rules to determine if packets are allowed to travel from one network to another. they restrict access from outside prevents attackers from getting too close and restrict people from leaving. To attain a certain level of security you can: regulate traffic allowed, protect the traffic using encryption, monitor traffic and hosts for "bad behaviour".

firewalls rules/proprieties:

- Least privilege
- Defense in depth
- choke point
- weakest links
- fail-safe stance
- universal partecipation
- diversity of defense
- simplicity

4.1.1 Host based packet filters

A kind of firewall that disciplines traffic in/out a single host, specifies the packets that can be received and sent. Usually each installed application has a known policy that has to obey

4.1.2 Network access

Bibliography